

Embedded Linux & System Software Interview Handbook

Chapter 7 - Linux Kernel Synchronization & Concurrency (Part 5)

Topics: Wait Queues, RCU, Deadlocks, Lockdep and Concurrency Debugging

7.9 Wait Queues

Wait queues allow a process to sleep until a specific condition becomes true. They are widely used in Linux device drivers to avoid busy waiting while waiting for hardware events or data.

Common Wait Queue APIs

API	Purpose
<code>init_waitqueue_head()</code>	Initialize a wait queue
<code>wait_event()</code>	Sleep until a condition is true
<code>wait_event_interruptible()</code>	Interruptible wait
<code>wake_up()</code>	Wake waiting tasks
<code>wake_up_interruptible()</code>	Wake interruptible tasks

7.10 Read-Copy-Update (RCU)

RCU is a high-performance synchronization mechanism optimized for workloads with many readers and few writers. Readers execute without blocking, while writers create updated copies and defer reclamation until all readers finish.

Typical RCU APIs

- `rcu_read_lock()`
- `rcu_read_unlock()`
- `synchronize_rcu()`
- `call_rcu()`

7.11 Deadlocks

A deadlock occurs when two or more execution contexts wait indefinitely for each other to release resources. Common causes include inconsistent lock ordering, forgotten unlocks and recursive locking.

Avoiding Deadlocks

- Acquire locks in a consistent order.
- Keep critical sections short.
- Avoid holding multiple locks unnecessarily.
- Release locks on every error path.

Lockdep and Kernel Debugging

Lockdep is the Linux kernel's lock dependency validator. It detects potential deadlocks, incorrect lock ordering and other synchronization bugs during development.

Useful Debugging Tools

Tool	Purpose
lockdep	Detect locking issues
ftrace	Trace kernel execution
dmesg	Kernel logs
/proc/locks	Active file locks
CONFIG_PROVE_LOCKING	Kernel lock validation
CONFIG_DEBUG_SPINLOCK	Spinlock debugging

Interview Scenarios

- A driver occasionally hangs under heavy load.
- Two kernel threads stop making progress.
- `wait_event()` never returns.
- RCU callback is never executed.
- lockdep reports a circular locking dependency.

Interview Questions

1. What is a wait queue?
2. Why use wait queues instead of polling?
3. Explain RCU and its advantages.
4. What causes deadlocks?
5. How does lockdep help kernel developers?
6. How would you debug a driver that hangs intermittently?

Key Takeaways

Wait queues, RCU and lock debugging tools are essential for building scalable and reliable Linux kernel software. Mastering these mechanisms enables developers to diagnose complex concurrency bugs in production systems and perform confidently in senior embedded Linux interviews.